

Elements of Electrical Engineering

Measuring Instruments	Principle and Construction of Instruments used for Measuring Current, Voltage, Power and Energy, Methods and precautions in use of these.
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Classification of Electrical Measuring Instruments.

The various electrical measuring instruments can be broadly divided into two classes (i) **Absolute instruments** (ii) **Secondary instruments**.

(i) Absolute instruments

Those instruments which give the value of the quantity to be measured in terms of the constants of the instrument and its deflection are called absolute instruments.

For example, a tangent galvanometer which is used to measure current is an absolute instrument.

- A tangent galvanometer gives the value of current being measured in terms of the tangent of the angle of deflection, the radius and number of turns of the coil and the horizontal component of earth's magnetic field.
- No previous calibration or comparison is necessary in case of absolute instruments.
- They are used only in standard laboratories as standardising instruments.

(ii) **Secondary instruments**

Those instruments in which the electrical quantity being measured is given directly by the deflection of the instrument are called *secondary instruments*.

- These instruments are provided with a calibrated scale.
- The calibration is done with the help of an absolute instrument or another calibrated instrument.
- These are the secondary instruments which are most generally used in everyday work.
- For example, ***they are very commonly used in laboratories, power stations, substations, industries etc.***

Types of Secondary Instruments

Secondary instruments may be classified according to their functions as

(i) indicating instruments (ii) integrating instruments and (iii) recording instruments.

(i) Indicating instruments.

Those instruments **which directly indicate the value of the electrical quantity at the time** when it is being measured are called *indicating instruments* e.g. **ammeters, voltmeters** and **wattmeters**.

- In such instruments, a pointer moving over a graduated scale directly gives the value of the electrical quantity being measured.
- For example, when an ammeter is connected in the circuit, the pointer of the meter directly indicates the value of current flowing in the circuit at that time.



(ii) **Integrating instruments.** Those instruments *which measure the total quantity of electricity* (in ampere-hours) or electrical energy (in watt-hours) in a given time are called *integrating instruments* e.g. **ampere-hour meter** and **watt-hour meter**.

In such instruments, there are sets of dials and pointers which register the total quantity of electricity or electrical energy supplied to the load.



(iii) Recording instruments Those instruments *which give a continuous record of the variations of the electrical quantity* to be measured are called *recording instruments*.

- A recording instrument is merely an indicating instrument with a pen attached to its pointer.
- The pen rests lightly on a chart wrapped over a drum moving with a slow uniform speed. The motion of the drum is in a direction perpendicular to the direction of the pointer. The path traced out by the pen indicates the manner in which the quantity, being measured, has varied during the time of the record. Recording voltmeters are used in supply stations to record the voltage of the supply mains during the day. Recording ammeters are employed in supply stations for registering the current taken from the batteries.



Principles of Operation of Electrical Instruments

❑ An electrical instrument essentially consists of a movable element and a scale to indicate or register the electrical quantity being measured.

❑ The movement of the movable element is caused by utilizing one or more of the following effects of current or voltage.

1. *Magnetic effect* Moving-iron instruments
2. *Electrodynamic effect* (i) Permanent-magnet moving coil
(ii) Dynamometer type
3. *Electromagnetic-induction* Induction type instruments
4. *Thermal effect* Hot-wire instruments
5. *Chemical effect* Electrolytic instruments
6. *Electrostatic effect* Electrostatic instruments

Essentials of Indicating Instruments

In order to ensure proper operation of indicating instruments, the following three torques are required :

(i) Deflecting (or operating) torque

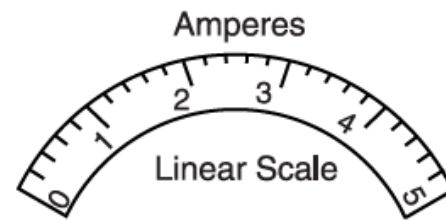
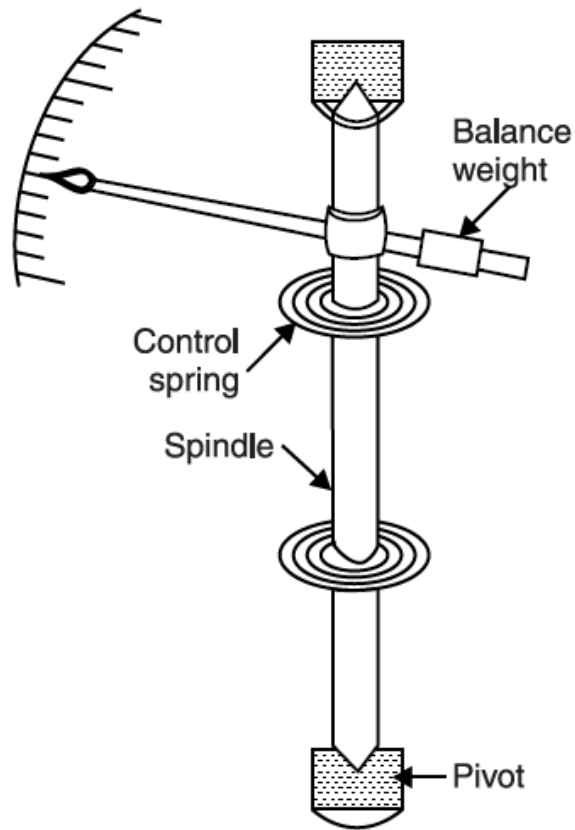
The deflecting torque is produced by utilizing the various effects of electric current or voltage and causes the moving system (and hence the pointer) to move from zero position.

(ii) Controlling (or restoring) torque

The controlling torque is provided by spring or gravity and opposes the deflecting torque. The pointer comes to rest at a position where these two opposing torques are equal.

(iii) Damping torque

The damping torque is provided by air friction or eddy currents. It ensures that the pointer comes to the final position without oscillations, thus enabling accurate and quick readings to be taken.



Controlling (or restoring) torque

The controlling torque performs two functions

- (i) It increases with the deflection of the moving system so that the final position of the pointer on the scale will be according to the magnitude of current (or voltage or power) to be measured.
- (ii) It brings the pointer back to zero position when the deflecting torque is removed. If it were not provided, *the pointer once deflected would not return to zero position* on removing the deflecting torque.

The controlling torque in indicating instruments may be provided by one of the following two methods :

1. By one or more springs ... *Spring control*
2. By weight of moving parts ... *Gravity control*

Spring Control

This is the most common method of providing controlling torque in electrical instruments.

- A spiral hairspring made of some non-magnetic material like phosphor bronze is attached to the moving system of the instrument. With the deflection of the pointer, the spring is twisted in the opposite direction. This twist in the spring provides the controlling torque.
- Since the torsion torque of a spiral spring is proportional to the angle of twist, the controlling torque is directly proportional to the deflection of the pointer *i.e.* $TC \propto \theta$.

The pointer will come to rest at a position where controlling torque TC is equal to the deflecting torque Td ,

$$Td = TC.$$

In an instrument where the deflecting torque is uniform, spring control provides a linear or evenly-spaced scale over the whole range.

For example, in a permanent-magnet moving coil instrument, the deflecting torque is directly proportional to the current flowing through the operating coil
i.e.

$$Td \propto I$$

With spring control, $TC \propto \theta$

In the final deflected position, $Td = TC$

$$\therefore \theta \propto I$$

Since the deflection is directly proportional to I , scale of such an instrument will be linear (uniform)

Advantages

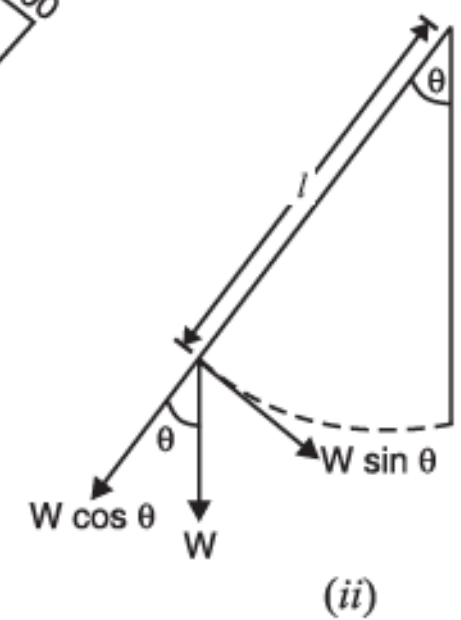
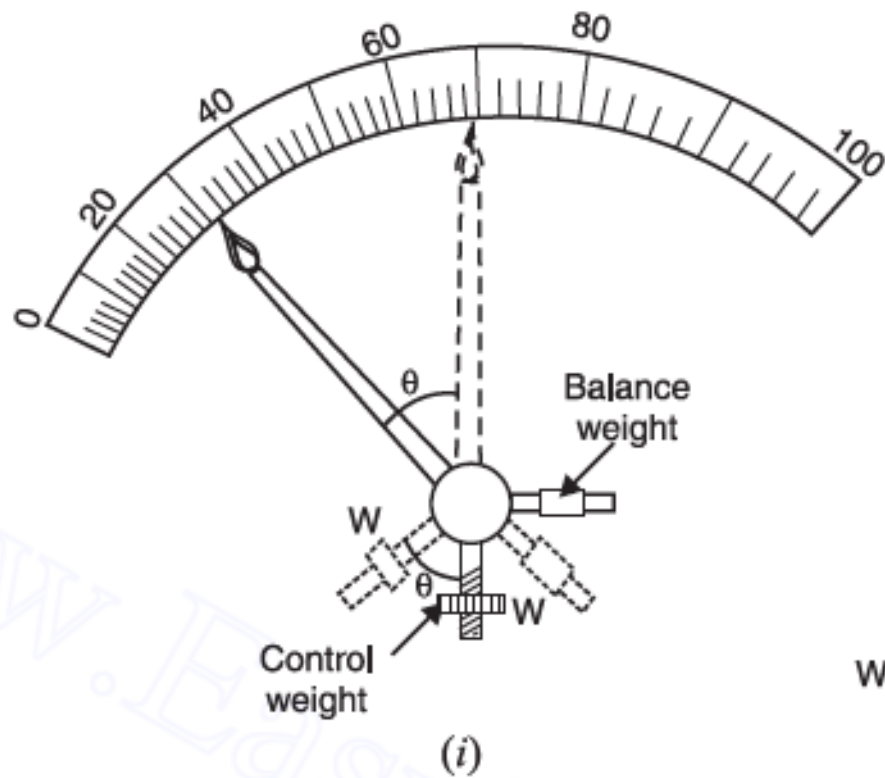
- (i) The levelling of the instrument is not required if the moving parts are balanced.
- (ii) In some instruments (*e.g.* permanent-magnet moving coil and dynamometer type), springs also serve as the current leads to the moving coil.
- (iii) There is practically no increase in the weight of the moving system.
- (iv) In instruments where deflecting torque is uniform, spring control provides uniform scale.

Disadvantages

- (i) Change of temperature affects the spring length and hence the controlling torque.
- (ii) Controlling torque cannot be adjusted easily.
- (iii) Accidental stresses in the springs may damage them.

2. Gravity Control.

- In this method, a small adjustable weight W is attached to the moving system which provides the necessary controlling torque.
- In the zero position of the pointer, the control weight hangs vertically downward and therefore provides no controlling torque.
- However, under the action of deflecting torque, the pointer moves from zero position (from left to right) and control weight moves in the opposite direction.
- Due to gravity, the control weight would tend to come to original position (*i.e.* vertical) and thus provides an opposing or controlling torque.
- The pointer comes to rest at a position where controlling torque is equal to the deflecting torque.



In the deflected position weight W can be resolved into two rectangular components *viz* $W \cos \theta$ and $W \sin \theta$. Only the component $W \sin \theta$ provides the controlling torque TC .

$$\therefore TC = Wl \sin \theta \quad \text{or } TC \propto \sin \theta \text{ (for fixed } W \text{ and } l)$$

Thus, the controlling torque is proportional to the sine of angle of deflection. It may be seen that the value of TC can be varied by changing l *i.e.* by changing the position of control weight W on the arm.

Taking the case of an instrument in which the deflecting torque is directly proportional to current (*e.g.* permanent-magnet moving coil instrument), we have for gravity control,

$$Td \propto I \text{ and } TC \propto \sin \theta$$

In the final deflected position, $Td = TC$.

$$\therefore I \propto \sin \theta$$

Since I is proportional to the sine of angle of deflection, gravity-controlled instruments have non-uniform scales ;

Advantages

- (i) It is slightly cheaper in manufacturing costs than spring control.
- (ii) It is unaffected by temperature variations.
- (iii) It is not subjected to fatigue.
- (iv) The controlling torque can be changed easily.

Disadvantages

- (i) The instrument has to be kept in vertical position.
- (ii) The control weight increases the weight of the moving system.
- (iii) Gravity-controlled instruments have non-uniform scale.

Damping torque

- (i) Due to the effect of deflecting torque (forward movement) and controlling torque (backward movement) and the inertia of motion the pointer (moving system in an instrument) is oscillate about its final position a number of times.
- (ii) So, for satisfactory and smooth operation damping torque is required, which move the pointer smoothly to the final reading position. Without a proper damping system, the instrument may be underdamp or overdamp one.

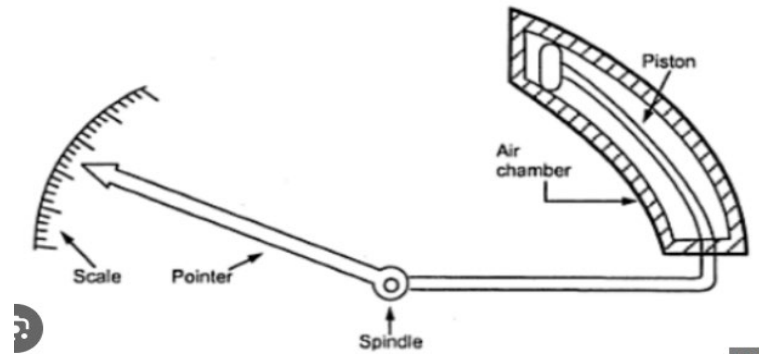
The controlling torque in indicating instruments may be provided by one of the following two methods :

The types of damping used are

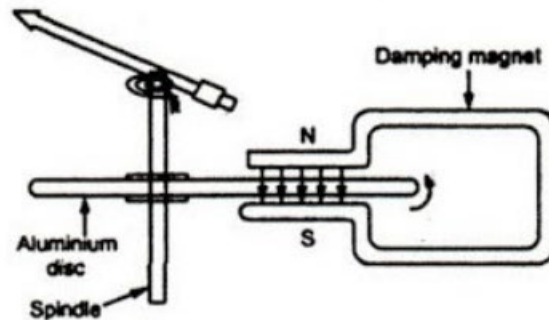
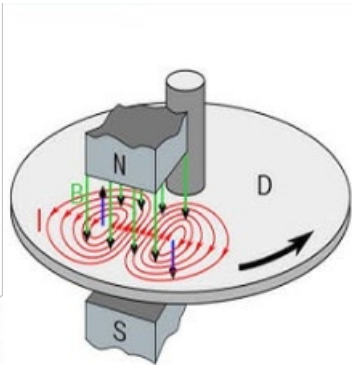
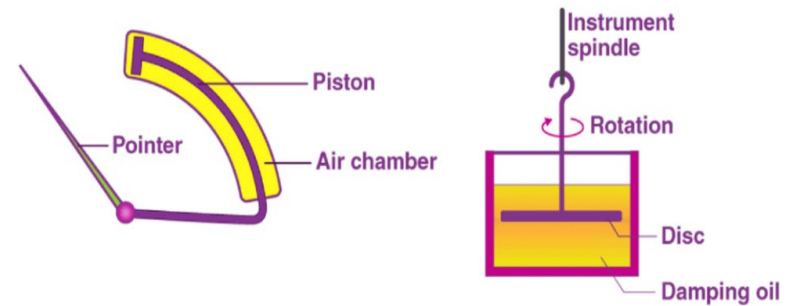
1. *Air friction Damping*
2. *Fluid friction damping*
3. *Eddy current Damping*

Damping torque

- (i) The damping is provided by using the flow of air inside a piston.
- (ii) The damping is provided by using the flow of fluid inside a piston.
- (iii) Damping is provided by the eddy current produced in the rotating disc.



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Types of instruments

The following types of instruments are used for making voltmeters and ammeters :

1. Permanent-magnet moving coil type (**PMMC**)
2. Dynamometer type
3. Moving-iron type
4. Hot-wire type
5. Electrostatic type (for voltmeter only)
6. Induction type

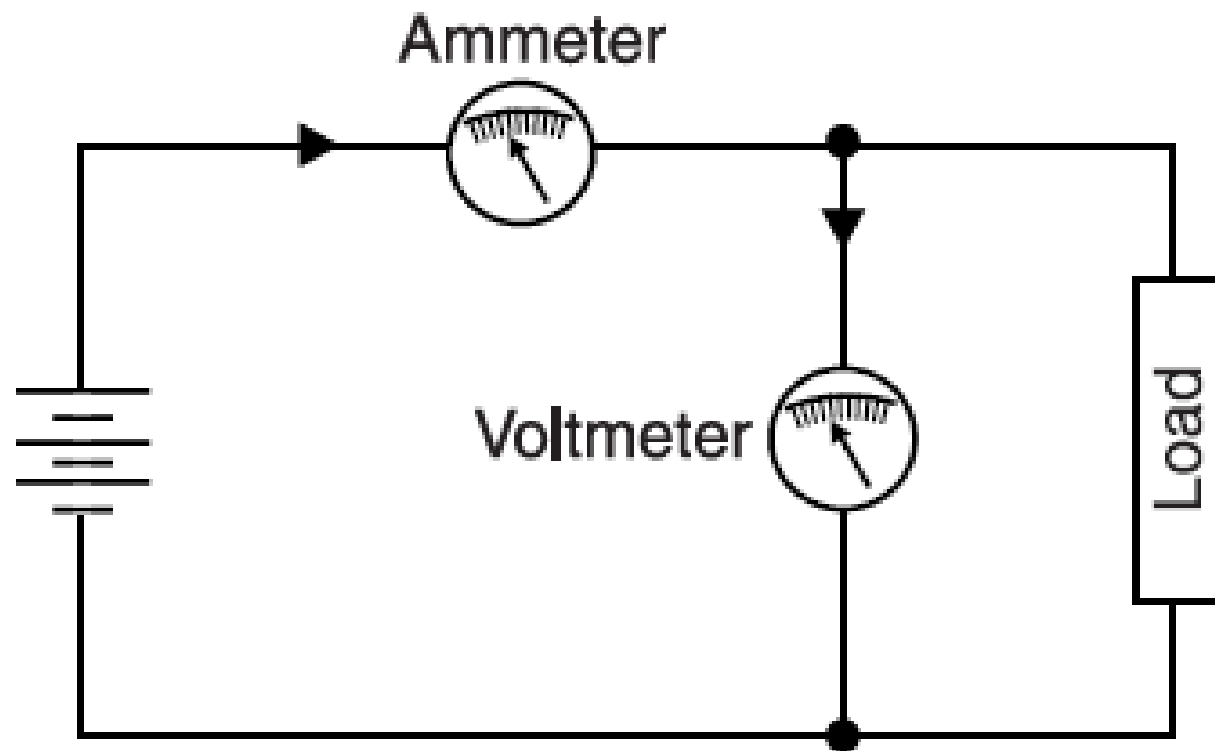
The instrument at Sr. No. (i) can be used for **DC** only whereas instrument at Sr. No. (vi) is employed for a.c. work only. However, instruments from Sr. No. (ii) to (v) can be used for both d.c. and a.c. measurements.

Ammeter

- ❑ An ammeter is used *to measure the flow of current* in a circuit.
- ❑ It is thus *connected in series* with the circuit under test so that current to be measured or a fraction of it passes through the instrument itself.
- ❑ The ammeter must be capable of carrying this current without injury to itself and without abnormally increasing the resistance of the circuit into which it is inserted.
- ❑ For this reason, *an ammeter is designed to have low resistance*.

Voltmeter

- ❑ A voltmeter is used to ***measure the potential difference*** between two points of a circuit.
- ❑ It is thus **connected in parallel** with the circuit or some part of the circuit.
- ❑ The voltmeter must have enough resistance so that it will not be injured by the current that flows through it, and so that it will not materially affect the current in the circuit to which it is connected.
- ❑ For this reason, **a voltmeter is designed to have high resistance.**



❑ *The basic principle of the ammeter and of the voltmeter is the same. Both are current operated devices i.e. **deflecting torque is produced when current flows through their operating coils.***

❑ In the ammeter, the deflecting torque is produced by the current that is measured, or a certain fraction of that current.

❑ In the voltmeter, the deflecting torque is produced by a current **which is proportional to the potential difference that is measured.**

❑ Thus, the same instrument can be used as an ammeter or voltmeter with proper design.

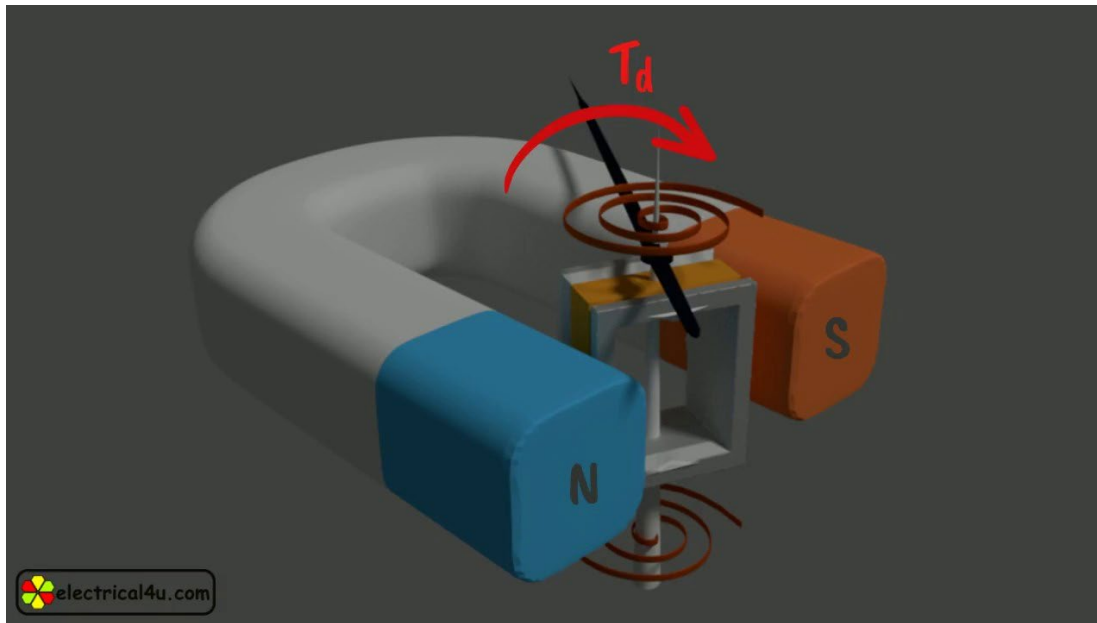
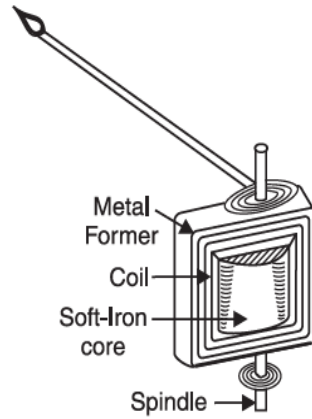
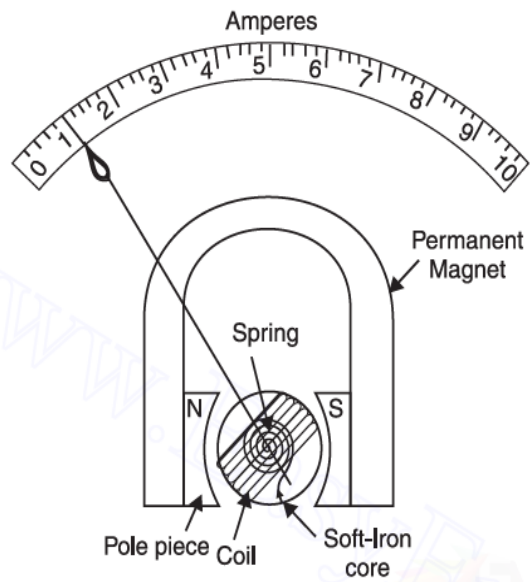
Permanent-Magnet Moving Coil (PMMC) Instruments (Ammeters and Voltmeters)

Construction

- ❑ These instruments are used either as ammeters or voltmeters and are suitable for dc only.
- ❑ *This type of instrument is based on the principle that “when a current carrying conductor is placed in a magnetic field, mechanical force acts on the conductor.”*
- ❑ The coil placed in the magnetic field and carrying the operating current is attached to the moving system.
- ❑ With the movement of the coil, the pointer moves over the scale to indicate the electrical quantity being measured.

Working

- When the instrument is connected in the circuit to measure current or voltage, the operating current flows through the coil. Since the **coil** is carrying current and is placed in the magnetic field of the **permanent magnet**, a mechanical torque acts on it.
- As a result, the **pointer** attached to the moving system moves in a clockwise direction over the graduated scale to indicate the value of current or voltage being measured.
- If the current in the coil is reversed, the deflecting torque will also be reversed since the direction of the field of the permanent magnet is the same. Consequently, the pointer will try to deflect below zero.
- Deflection in this direction (*i.e.* reverse direction) is prevented by a **spring** “stop”. *Since the deflecting torque reverses with the reversal of current in the coil, such instruments can be used to measure direct currents and voltages only.*



<https://www.youtube.com/watch?v=ZtBKC6WSjD0>

Deflecting torque

- ❑ The magnetic field in the air gap is radial due to the presence of soft-iron core.
- ❑ This means that conductors of the coil will always move at right angles to the field.
- ❑ When current is passed through the coil, ***forces act on its both sides which produce the deflecting torque.***

Let,

B = flux density in Wb/m²

l = length or depth of coil in m

b = breadth of coil in m

N = No. of turns in the coil

If a current of I amperes flows in the coil, then force acting on each coil side is given by

Force on each coil, $F = B I L N$ newtons

Deflecting torque, $Td = \text{Force} \times \text{perpendicular distance}$
 $= (B I L N) \times b$

$\therefore Td = B I N A$ newton-metre

where $A (= b \times L)$ is the area of the coil in m^2 . Since the values of B , N and A are fixed,

$\therefore Td \propto I$

The instrument is spring-controlled so that $TC \propto \theta$.

The pointer will come to rest at a position where

$Td = TC$.

$\therefore \theta \propto I$

Thus, the **deflection is directly proportional to the operating current.**
Hence, such instruments have uniform scale.

Advantages

- (i) Uniform scale *i.e.*, evenly divided scale.
- (ii) Very effective eddy current damping because the aluminium former moves in an intense magnetic field of the permanent magnet.
- (iii) High efficiency as it requires very little power for its operation.
- (iv) No hysteresis loss as the magnetic flux is practically constant.
- (v) External stray fields have little effect on the readings as the operating magnetic field is very strong.
- (vi) Very accurate and reliable.

Disadvantages

- (i) Such instruments cannot be used for ac measurements.
- (ii) About 50% more expensive than moving-iron instruments because of their accurate design.
- (iii) Some errors are caused due to variations (with time or temperature) either in the strength of permanent magnet or in the control springs.

❑ ***PMMC instruments are more suitable for all dc measurements.***

❑ They are very sensitive and maintain a high degree of accuracy over long periods.

The **major applications** of such instruments are

(i) In the measurement of direct currents and voltages.

(ii) In dc galvanometers to detect small currents.

(iii) In ballistic galvanometers used mainly for measuring changes of magnetic flux linkages.

Moving Iron (MI) Instruments (Ammeters and Voltmeters)

This type of instrument is principally used for the **measurement of alternating currents and voltages**, though it can also be used for dc measurements.

There are two types of moving-iron instruments

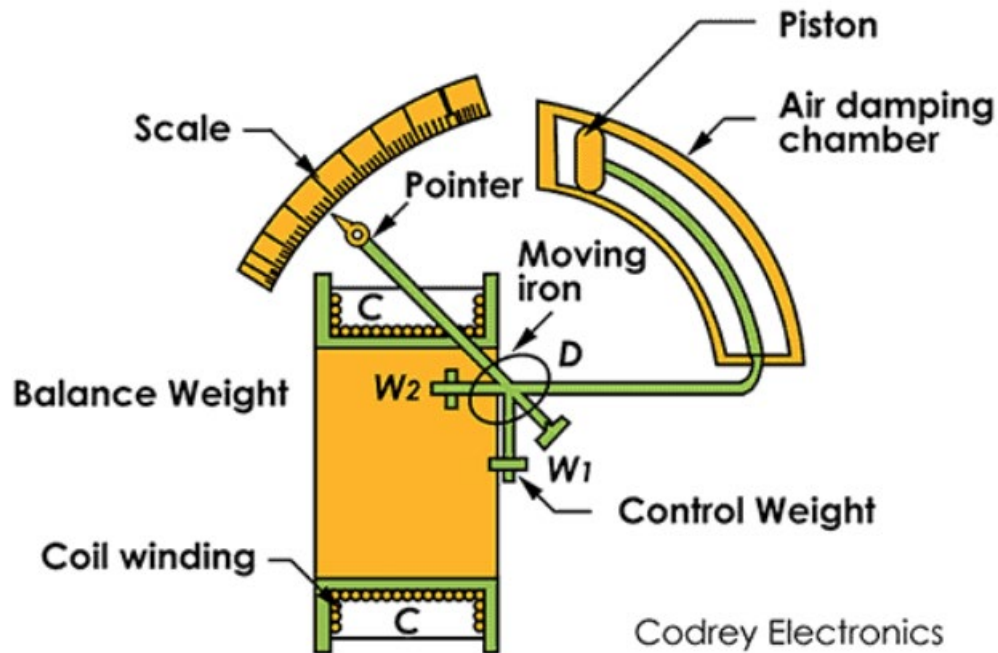
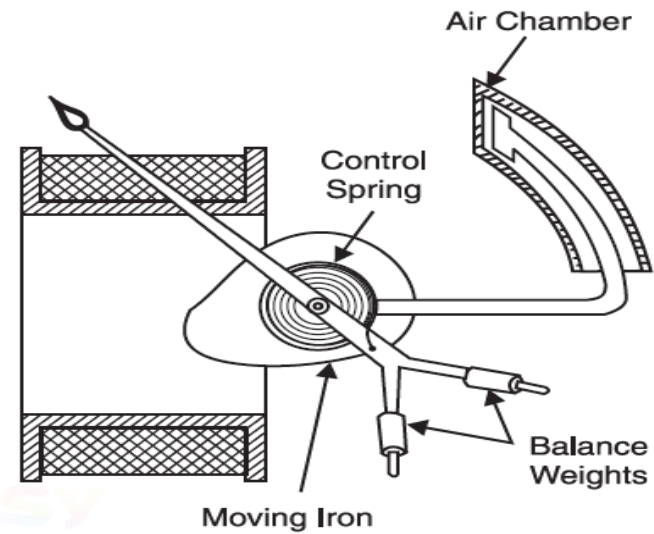
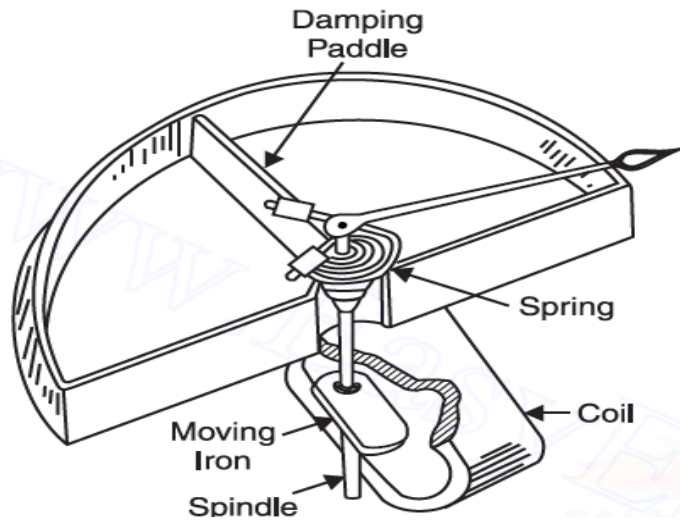
(i) Attraction type in which a single soft-iron vane (or moving iron) is mounted on the spindle and is attracted towards the coil when operating current flows through it.

(ii) Repulsion type in which two soft-iron vanes are used; one fixed and attached to the stationary coil while the other is movable (*i.e.* moving iron) and mounted on the spindle of the instrument.

- When operating current flows through the coil, the two vanes are magnetized, developing similar polarity at the same ends.
- Consequently, *repulsion takes place between the vanes and the movable vane causes the pointer to move over the scale.*
- In case the instrument is to be used as an ammeter, the coil has a fewer turns of thick wire so that the ammeter has low resistance – a desirable requirement.
- In case it is to be used as a voltmeter, the coil has a large number of turns of fine wire so that the voltmeter has high resistance – a desirable requirement.

Attraction Type M.I. Instruments

- It consists of a cylindrical coil or solenoid which is kept fixed.
- An oval-shaped soft-iron is attached to the spindle in such a way that it can move in and out of the coil.
- *A pointer is attached to the spindle so that it is deflected with the motion of the soft-iron piece.*
- The *controlling torque is provided by one spiral spring arranged at the top of the moving element.*
- It should be noted that in this instrument, the springs do not carry the current as the same is carried by the stationary coil.
- The damping device is an aluminium vane attached to the spindle, which moves in a closed chamber.
- In some instruments, damping is provided by the movement of a piston inside the curved chamber the piston being attached to the spindle.



Codrey Electronics

Deflecting torque - The force F pulling the soft-iron piece towards the coil is directly proportional to :

(i) magnetic field strength H produced by the coil

(ii) pole strength m developed in the iron piece

$$\begin{aligned}\therefore F &\propto mH \\ &\propto H^2 \quad (\because m \propto H)\end{aligned}$$

$$\therefore \text{Instantaneous deflecting torque} \propto H^2$$

If the permeability of iron is assumed constant, then,

$$H \propto i \quad \text{where } i \text{ is the instantaneous coil current.}$$

$$\therefore \text{Instantaneous deflecting torque} \propto i^2$$

Average deflecting torque, $T_d \propto \text{mean of } i^2 \text{ over a cycle}$

Since the instrument is spring controlled,

$$TC \propto \theta$$

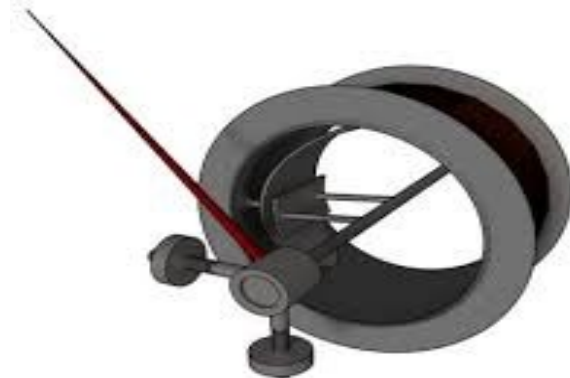
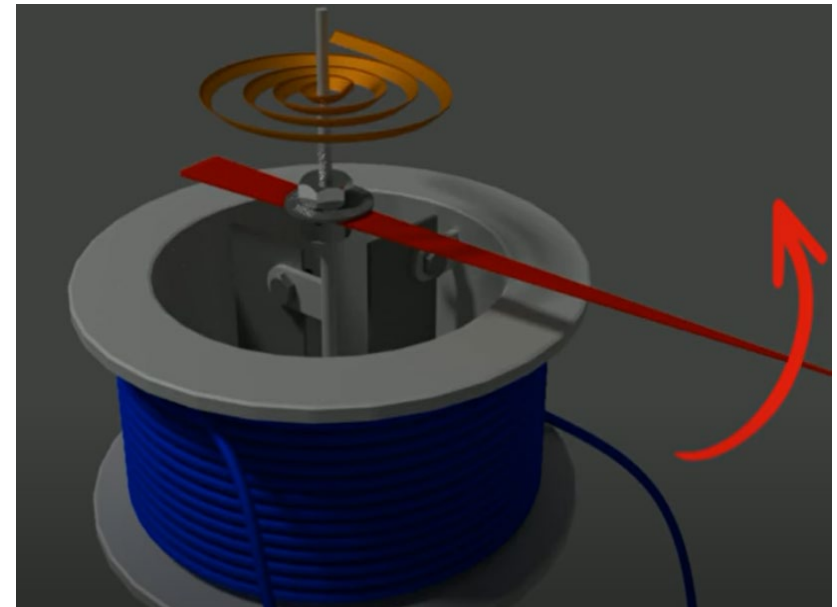
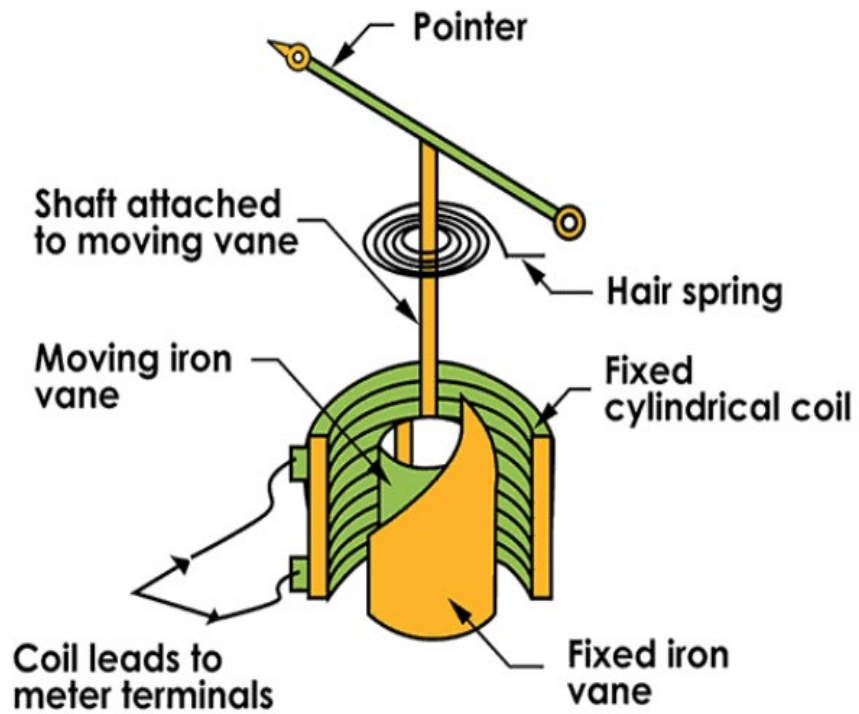
In the steady position of deflection, $Td = TC$.

$$\begin{aligned}\theta &\propto \text{mean of } \tilde{i}^2 \text{ over a cycle} \\ &\propto I^2 && \dots \text{for } d.c. \\ &\propto I_{r.m.s.}^2 && \dots \text{for } a.c.\end{aligned}$$

Since the deflection is proportional to the square of coil current, the scale of such instruments is non-uniform; being crowded in the beginning and spread out near the finish end of the scale.

Repulsion Type M.I. Instruments

- It consists of two soft-iron pieces or vanes surrounded by a fixed cylindrical hollow coil which carries the operating current. One of these vanes is fixed and the other is free to move.
- The movable vane is of cylindrical shape and is mounted axially on a spindle to which a pointer is attached. The fixed vane, which is wedge-shaped and has a larger radius, is attached to the stationary coil.
- The controlling torque is provided by one spiral spring at the top of the instrument.
- It may be noted that in this instrument, springs do not provide the electrical connections. Damping is provided by air friction due to the motion of a piston in an air chamber.



<https://www.youtube.com/watch?v=bYGgMvXMJMo&t=217s>

Working

- When current to be measured or current proportional to the voltage to be measured flows through the coil, a magnetic field is set up by the coil.
- This magnetic field magnetizes the two vanes in the same direction *i.e.*, similar polarities are developed at the same ends of the vanes.
- Since the adjacent edges of the vanes are of the same polarity, the two vanes repel each other. As the fixed vane cannot move, the movable vane deflects and causes the pointer to move from zero position.
- *The pointer will come to rest at a position where deflecting torque is equal to the controlling torque provided by the spring.*
- If the current in the coil is reversed, the direction of deflecting torque remains unchanged. It is because reversal of the field of the coil reverses the magnetization of both iron vanes so that they repel each other regardless of which way the current flows through the coil.
- For this reason, such instruments can be used for both dc and ac applications.

Deflecting torque

❑ The deflecting torque results due to the *repulsion between the similarly charged soft-iron pieces or vanes*.

If the two pieces developed pole strengths of m_1 and m_2 respectively, then,

Instantaneous deflecting torque $\propto m_1 m_2 \propto H^2$

If the permeability of iron is assumed constant, then

$H \propto i$ where i is coil current.

Instantaneous deflecting torque $\propto i^2$

$\therefore$ Average deflecting torque, $T_d \propto$ mean of i^2 over a cycle

Since the instrument is spring-controlled,

$\therefore TC \propto \theta$

In the steady position of deflection, $T_d = TC$.

$\therefore \theta \propto \text{mean of } i^2 \text{ over a cycle}$

$$\propto I^2 \text{ ...for dc}$$

$$\propto I^2 (\text{rms}) \text{ ... for ac}$$

Thus, the deflection is proportional to the square of coil current as is the case with attraction type moving-iron instruments.

❑ Therefore, the scale of such instruments is also non-uniform; being crowded in the beginning and spread out near the finish end of the scale.

❑ However, the non-linearity of the scale can be corrected to some extent by the accurate shaping (e.g., using tongue-shaped vanes) and positioning of iron vanes in relation to the operating coil.

Sources of Errors in Moving Iron Instruments

(1) errors with both dc and ac work (2) errors with ac work only.

1. Errors with both dc and ac work

(i) **Errors due to hysteresis.** Since the iron parts move in the magnetic field, hysteresis loss occurs in them. The effect of this error is that the readings are higher when current increases than when it decreases. This error can be reduced by employing vanes of *Mu-metal* and by working it over a low range of flux densities. (*Mu-metal* is a magnetic material having high permeability and low hysteresis loss.)

(ii) **Error due to stray fields.** Since the operating field is comparatively weak (say 7×10^{-3} Wb/m²), such instruments are readily affected by stray fields. This may give rise to wrong readings. This error can be reduced by enclosing the movement in an iron case.

(iii) **Error due to temperature.** Change of temperature affects the instrument resistance and stiffness of the control spring.

(iv) **Error due to friction.** Due to the friction of moving parts, a slight error may be introduced. This can be avoided by making torque-weight ratio high.

2. Errors with ac only.

With the change in frequency, the impedance of the instrument coil changes. This will cause a change of current in the coil. This is particularly important in case of voltmeters since these are connected in parallel with the circuit. The indicated voltage V is given by:

$$V = I_m \sqrt{(R_m + R_s)^2 + \omega^2 L_m^2}$$

where I_m is the meter current, R_m and L_m are the resistance and inductance of the coil of the meter and R_s is multiplier. For high frequencies, the meter gives low readings and *vice-versa*. This error can be eliminated by connecting a capacitor of suitable value in parallel with the series resistance R_s of the voltmeter. The value of capacitor is given by $C = L_m/R_s$

2.

Characteristics of Moving-Iron Instruments

Advantages

- (i) These are less expensive, robust and simple in construction.
- (ii) These can be used for both dc and ac measurements. However, when used with dc , they are liable to small errors due to residual magnetism.
- (iii) These instruments have high operating torque.
- (iv) These instruments are reasonably accurate.

Disadvantages

- (i) These instruments have non-linear scales.
- (ii) These instruments are not as sensitive as the PMMC instruments.
- (iii) Errors are introduced due to change in frequency in case of ac measurements.

Applications

- ✓ *The moving-iron instruments are **primarily used for ac measurements** .*
- ✓ They are not used to measure direct currents and voltages because their characteristics are inferior to permanent-magnet moving coil instruments.

Wattmeter

*A wattmeter **measures electric power** given to or developed by an electric apparatus or circuit.*

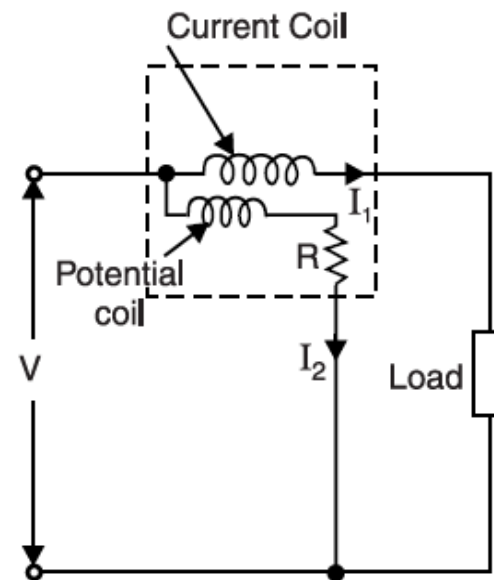
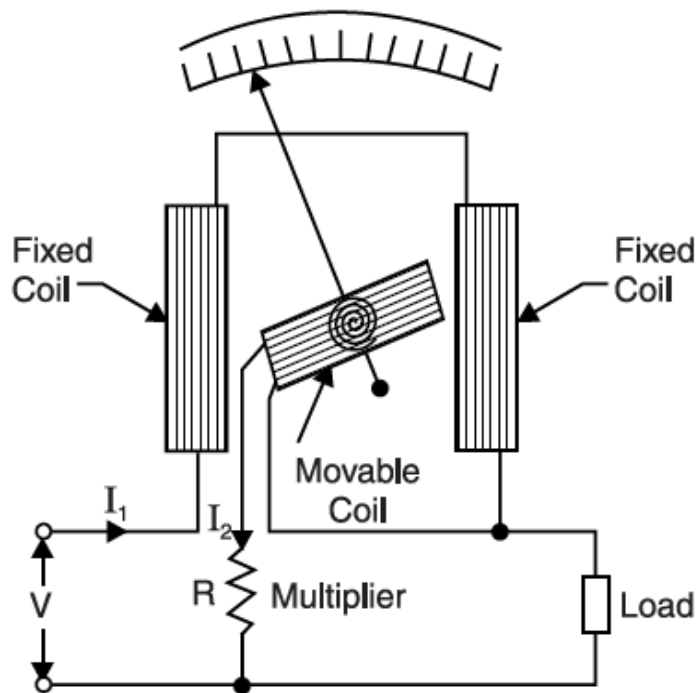
- A wattmeter is hardly ever required in a dc circuit because power ($P = VI$) can be easily determined from voltmeter and ammeter readings.
- However, in an a.c circuit, power ($P = VI \cos \phi$) depends not only on voltage and current but also on the phase shift between them.

Therefore, a wattmeter is necessary for a.c. power measurement. There are two principal types of wattmeters,

- (i) Dynamometer wattmeter — for both d.c. and a.c. power
- (ii) Induction wattmeter — for a.c. power only

Dynamometer Wattmeter

- ❑ A dynamometer wattmeter is almost universally used for the *measurement of dc as well as ac power*.
- ❑ It works on the dynamometer principle i.e., *mechanical force exists between two current carrying conductors or coils*.



Characteristics of Dynamometer Wattmeter

It is worthwhile to give the advantages, disadvantages, errors, ratings and range extension of dynamometer wattmeters.

Advantages

- (i) They can be used for the measurement of a.c. as well as d.c. power.
- (ii) They have uniform scale unlike dynamometer ammeters and voltmeters which have square law scales.
- (iii) By careful design, high accuracy can be obtained.

Disadvantages

- (i) At low power factors, the inductance of the potential coil causes serious errors.
- (ii) The readings of the instrument may be affected by stray magnetic fields. In order to prevent it, the instrument is shielded from the external magnetic fields by enclosing it in a soft-iron case.

Wattmeter ratings

- ❑ A wattmeter has two circuits - **current coil circuit and potential coil circuit**.
- ❑ These circuits (fixed coils and the movable coil) may burn out on carrying too much current.
- ❑ For this reason, current and voltage ratings are marked on the instrument and care should be taken that neither of these limits is exceeded.
- ❑ It has been observed that coils of wattmeter are burnt up while their readings stood much less than full-scale value.
- ❑ This particularly happens when the load power factor is low. ***At low power factors, the current through the current coil may exceed the safe value***, thus burning the current coils of the instrument.
- ❑ Because a wattmeter gives no indication of current and voltage values, an ammeter and a voltmeter should always be used in conjunction with wattmeter when the load p.f. is suspected to be low or whenever the load power factor is not known.

Errors

A wattmeter may not give true reading due to several sources of errors such as

- (i) error due to connection of potential coil circuit
- (ii) error due to inductance of potential coil
- (iii) error due to capacitance in potential coil circuit
- (iv) error due to stray fields and
- (i) error due to eddy currents.

S.No.	Particular	Moving coil	Dynamometer type	Moving iron
1.	<i>Construction</i>	Delicate construction	Heavy moving system	Simple
2.	<i>Cost</i>	Very high	High	Low
3.	<i>Power consumption</i>	Very low	High	Less than dynamometer type
4	<i>Scale</i>	Uniform	Non-uniform	Non-uniform
5	<i>Torque-weight ratio</i>	High	Small	More than dynamometer type
6.	<i>Suitable for</i>	d.c. only	d.c. as well as a.c.	d.c. as well as a.c.
7.	<i>Effect of stray magnetic fields</i>	Not affected	Affected	Not affected
8.	<i>Application</i>	Voltmeters and Ammeters	Generally for wattmeters	Voltmeters and ammeters
9.	<i>Accuracy</i>	High	Poor	Reasonable
10	<i>Sensitivity</i>	High	Poor	Reasonable